

Figure 1: **RQ1.** Paired accuracy on 150 eliciture pairs with Wilson 95% CIs. Dashed line: chance.

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Abstract

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1 Introduction

We ask three questions:

RQ1 *Do model behaviors reflect inferences about implicit causal relationships within sentences?*
RQ2 *Do LMs form internal representations of implicit causal relationships?*
RQ3 *If **RQ2** is true, do representations of implicit causality generalize to naturalistic settings with potential real-world implications?*

Our contribution is...

2 Background

3 Experiments

3.1 Study 1: Measuring Behavioral Eliciture

We vary the response measure in a conceptual replication of Pan and Kehler, finding substantial measurement dependence.

Methods. **Models.** We evaluate the base checkpoints of Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024].

Dataset and analysis. We evaluate 150 critical/control pairs with yes/no judgments, expected seven-point Likert scores, and the relative probability of the same continuation after *because* versus bare *and*. The last measure uses no metalinguistic prompt. A pair is correct when the critical item receives the stronger causal response, allowing us to compare prompting with probability measurement [Hu and Levy, 2023].

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Results. Figure 1 shows that Qwen3 4B/8B reach 1.00 with Likert responses and at least .96 with yes/no, but fall to .79/.76 under probability assignment. Conversely, Gemma 2 2B scores .45/.39 with prompting but .83 with *because* versus *and*. Overall, no interface supports a model-independent conclusion.

3.2 Study 2: Relation-Selective Causal Representations

We compare cross-sentence causal directions with temporal and plausibility directions, finding strong diagonals and weaker cross-transfer.

Methods. **Datasets.** Causal, temporal, and plausibility directions respectively use 384/96, 800/200, and 56/14 train/ID-development pairs.

Analysis. At every fourth layer and the final layer, CAA averages final-token differences between matched positive and negative texts. We select the earliest layer maximizing mean own-development paired accuracy and ROC–AUC, then apply that frozen direction to every target relation. Matrix diagonals measure within-relation discrimination; off-diagonals measure cross-transfer. This is an ID-development analysis, not an OOD test.

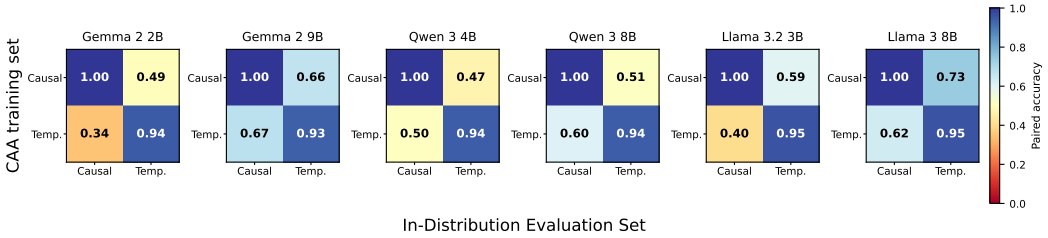


Figure 2: **RQ2a, causal vs. temporal.** ID-development paired accuracy when each row's CAA direction is evaluated on each column's relation. Each panel is one model; diagonals measure own-relation discrimination and off-diagonals measure cross-transfer.

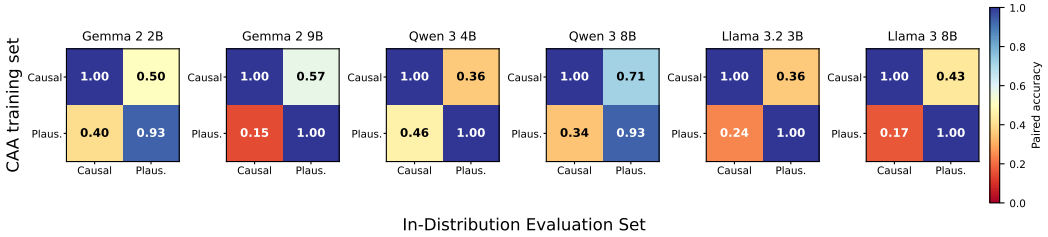


Figure 3: **RQ2a, causal vs. plausibility.** Six-model cross-transfer matrices; plausibility cells ($n = 14$ pairs) are exploratory.

Results. Figures 2–3 show causal self-transfer of 1.00 for all six models, temporal self-transfer of .93–.95, and plausibility self-transfer of .93–1.00. However, causal-to-temporal accuracy is .47–.73 and causal-to-plausibility is .36–.71; reverse transfer reaches .34–.67 and .15–.46. The gaps support relation selectivity, while occasional above-chance transfer argues against strict orthogonality.

3.3 Study 3: Generalization to Causal OOD

We next test whether causal directions remain selectively informative on unseen causal scenarios.

Methods. **Datasets.** Causal and temporal CAAs use 1,000/200 and 800/200 train/development pairs; the plausibility CAA uses 70 probable/impossible pairs with five-fold layer selection. We evaluate every frozen direction on the same 150-pair causal OOD set.

Analysis. We project each critical/control sentence onto the corresponding direction. A pair is correct when the critical item projects farther than its matched control.

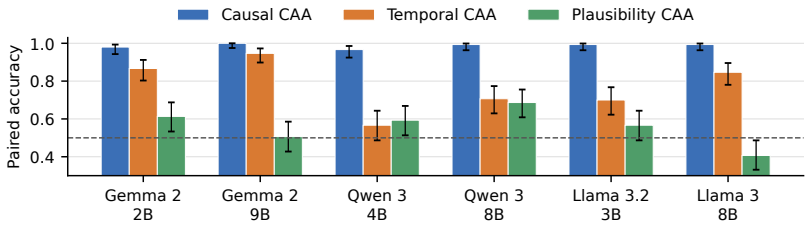


Figure 4: **RQ2b.** Paired accuracy on 150 causal OOD pairs with Wilson 95% CIs. Dashed line: chance.

Results. Figure 4 shows that causal CAA reaches .97–1.00 paired accuracy across models. Temporal CAA varies from .57–.95, whereas plausibility CAA ranges from .41–.69. The causal direction therefore generalizes most consistently to new causal scenarios; transfer from the comparison directions is weaker and model-dependent.

3.4 Study 4: Generalization to Naturalistic Scenarios

We finally test causal representations on matched naturalistic scenarios.

Methods. Dataset. We evaluate 65 self-blinding scenarios per model in matched White-male, minority-male, and minority-female versions.

Analysis. A unit causal direction fit separately on 1,000 within-sentence pairs projects each scenario's final-token state. We report within-scenario differences from the White-male version with 10,000-sample bootstrap 95% intervals and visualize raw, unnormalized CAA projections.

Table 1: **RQ3.** Mean paired unit-CAA change ± bootstrap 95% CI half-width (65 scenarios/model).

	Gemma 2 2B	Gemma 2 9B	Qwen3 4B	Qwen3 8B	Llama 3.2 3B	Llama 3 8B
Minority male − White male	**.41 ± .32**	**1.9 ± .40**	.073 ± .093	**.29 ± .17**	.025 ± .029	**.030 ± .020**
Minority female − White male	**.57 ± .32**	**2.3 ± .41**	**.15 ± .10**	**.39 ± .15**	**.069 ± .033**	**.060 ± .021**

Results. Table 1 shows positive estimates for both contrasts in every model. Minority-female intervals exclude zero for all six; minority-male intervals do so for four. However, Appendix Figure 5 shows substantial distributional overlap. These associations do not establish social bias, behavior, or a causal mechanism.

4 Discussion

Behavioral evidence is measurement-sensitive, whereas causal directions are relation-selective and generalize to causal OOD data. However, Study 2 uses ID development sets (only 14 plausibility pairs), CAA is correlational, and 65 scenarios cannot establish general behavior. Larger tests, interventions, and behavioral validation remain necessary.

References

Abhimanyu Dubey et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL `https://arxiv.org/abs/2407.21783`.

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A Raw CAA Projection Distributions

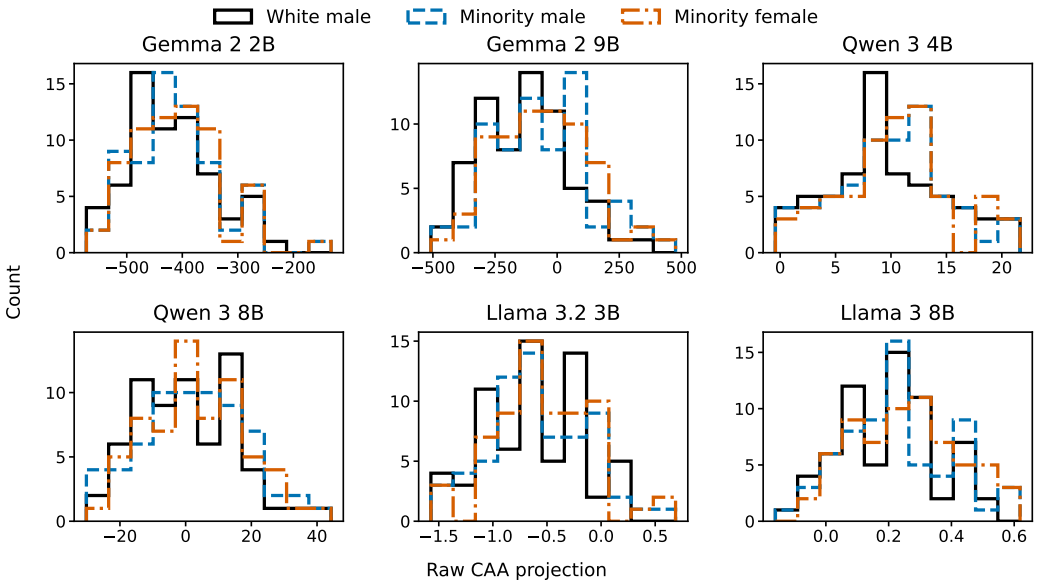


Figure 5: **RQ3.** Raw CAA projections by matched condition (65 scores/condition/model). Colors and line styles distinguish the three conditions.